Properties of Materials for Prestressed concrete and Flexural Behaviour of Uncracked members

Week 2

Professor Rebecca Gravina

The University of Queensland
Australia

Contents

•**Chapters 2 & 4 Prestressed Concrete, *Warner Foster Gravina* (Reference book)**

PROPERTIES OF
MATERIALS USED IN PC STRUCTURES

Introduction to Materials Used in PC Structures

- The properties of various materials used in PC structures will be introduced. Relevant clauses in AS 3600-2018 will be referred to explain the allowable design limits of the strength of materials.

- Time-dependent long-term effect on will be explained. Formulas specified in AS 3600-2018 will be introduced.

- The following will be discussed:
  (1) Concrete
  (2) Prestressing wires/strands
  (3) Untensioned (non-prestressing reinforcement)

- Short-term and Long-term properties of these materials will also be discussed.

Introduction to Materials Used in PC Structures

- The short-term properties include:
  - strength (compressive and/or tensile)
  - modulus of elasticity
  - stress-strain curve

- The long-term properties include:
  (1) Shrinkage of concrete
  (2) Creep of concrete
  (3) Relaxation of prestressing wires/strands

  However, relaxation of non-prestressing steel is neglected.

CONCRETE

Concrete

Characteristic strength

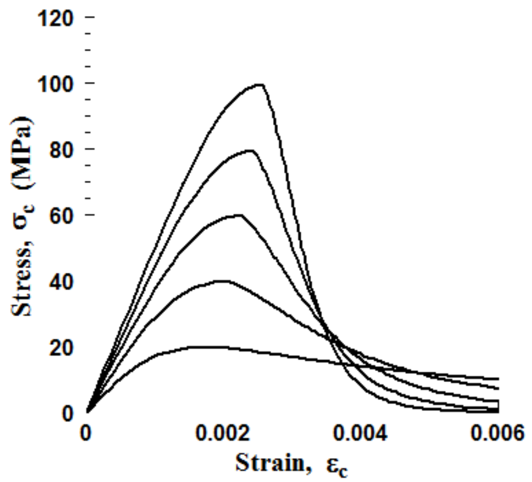
- The characteristic strength (compressive or tensile) of concrete is normally used in design. It is generally accepted that only 5% of the concrete samples tested under standard procedure will have strength lower than the characteristic strength.

- The characteristic compressive strength of concrete at 28 days (f_c') shall either be:

  - Equal to the specified strength grade, provided the appropriate curing is ensured and that the concrete complies with AS 1379;

  - Determined statistically from compressive strength tests carried out in accordance with AS1012.9.

- Standard values of f_c' are 20, 25, 32, 40, 50, 65, 80 and 100 MPa.



Concrete

- Table 3.1.2 of AS 3600-2018

TABLE 3.1.2

CONCRETE PROPERTIES AT 28 DAYS

f'_c MPa	20	25	32	40	50	65	80	100	120
f_{cm} MPa	25	31	39	48	59	75	91	110	128
f_{cmi} MPa	22	28	35	43	53	68	82	99	115
E_c MPa	24 000	26 700	30 100	32 800	34 800	37 400	39 600	42 200	44 400

- The flexural tensile strength ($f_{ct,f}$) and uni-axial tensile strength (f_{ct}) of concrete can be determined either from the flexural tensile strength of concrete or the characteristic uni-axial tensile strength of concrete.

- However, in the absence of more accurate data, the characteristic flexural and uni-axial tensile strength of concrete can be taken as $f'_{ct,f} = 0.6\sqrt{f'_c}$ and $f'_{ct} \approx 0.36\sqrt{f'_c}$ respectively.

Concrete

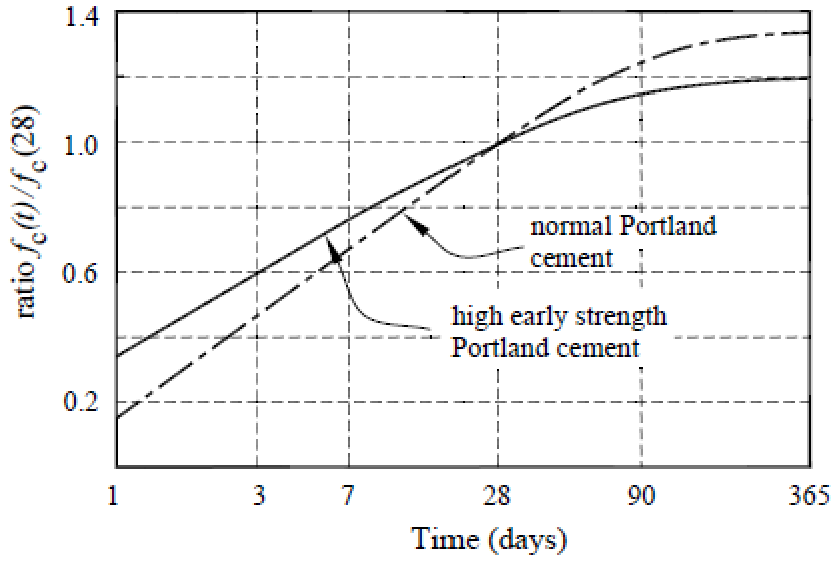
Stress limits for designing PC structures:

- Only one guideline stipulated in AS 3600-2018 for the stress limit of designing prestressed concrete structures, which is in Clause 8.1.6.2.

- The maximum compressive stress in concrete, under load transfer, should not exceed $0.5f_{cp}'$, where f_{cp}' is the characteristic compressive cylinder strength at transfer.

- Designers can also decide what values of permissible stresses to be used at transfer and service stages. For example, maximum tensile stress in concrete at transfer $\leq 0.5f_{ct,f}'$ (characteristic flexural tensile strength at transfer, not at 28 days)

Concrete



- Normally, load transfer can take place between 4 to 7 days after concreting (or longer). The strength of concrete (immature) will be lower than the 28th day strength (mature). It is about 40 – 70% depending on the curing condition.

Concrete

- Normally, load transfer can take place between 4 to 7 days after concreting (or longer). The strength of concrete (immature) will be lower than the 28th day strength (mature). It is about 40 – 80% depending on the curing condition.

- Say f_c' = 25 and 50 MPa at transfer and service respectively. Also, assume that $f_{ct,f}' = 0.6\sqrt{f_c'}$. Therefore, the permissible tensile strength can be taken as $0.3\sqrt{f_c'}$.

Concrete

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Elastic modulus:

- The mean modulus of elasticity of concrete at the appropriate age (E_{cj}) shall be following the formulas specified in Clause 3.1.2 of AS 3600-2018. Those formulas are listed below:

$$E_{cj} = \rho^{1.5}\left(0.043\sqrt{f_{cmi}}\right) \text{ [MPa] for } f_{cmi} \leq 40 \text{ MPa}$$

$$E_{cj} = \rho^{1.5}\left(0.024\sqrt{f_{cmi}} + 0.12\right) \text{ [MPa] for } f_{cmi} \geq 40 \text{ MPa}$$

- ρ is the density of concrete, which can be taken as 2,400 kg/m^3 as stipulated in Clause 3.1.3 of AS 3600-2018, and f_{cmi} is the mean in-situ compressive strength of concrete [MPa], the value of which can be determined by Clause 3.1.1.2 or Table 3.1.2.

Concrete

- Again, given f_c' at transfer and service are 25 and 50 MPa respectively, the value of f_{cmi} can be determined as 28 and 53 MPa respectively (Table 3.1.2). Therefore, the elastic moduli are respectively:

$$E_{cj} = 2400^{1.5}\left(0.043\sqrt{28}\right) = 26752 \text{ MPa}$$

$$E_{cj} = 2400^{1.5}\left(0.024\sqrt{53} + 0.12\right) = 34652 \text{ MPa}$$

- The elastic moduli of concrete can also be determined from Table 3.1.2. Intermediate concrete strength can be determined by interpolation. Small difference will be obtained by using table and equation.

- The elastic modulus of concrete is useful in estimating the short-term prestress loss due to elastic shortening and anchorage draw-in, as well as the long-term loss due to creep of concrete.

Concrete

Shrinkage:

- With loading or not, concrete contracts on drying and undergoes shrinkage.

- Two types of shrinkage:
  (1) Autogenous shrinkage – shrinkage of hydrated cement paste in a conservative system;
  (2) Drying shrinkage – withdrawal of water from concrete stored in unsaturated air.

- The designs shrinkage strain of concrete (ε_{cs}) can be determined as the sum of the chemical (autogenous) shrinkage strain (ε_{cse}) and the drying shrinkage strain (ε_{csd}) as per Clause 3.1.7.2 of AS 3600-2018.

Concrete

- The equations are:

Total shrinkage strain: $\varepsilon_{cs} = \varepsilon_{cse} + \varepsilon_{csd}$

Autogenous shrinkage strain:

$$\varepsilon_{cse} = \varepsilon^*_{cse}(1.0 - e^{-0.07t})$$

Final autogenous shrinkage strain (fc'≤ 50 and > 50 MPa):

$$\varepsilon^*_{cse} = 50 \times 10^{-6}(0.07f'_c - \mathbf{0.5})$$

$$\varepsilon^*_{cse} = 50 \times 10^{-6}(0.08f'_c - \mathbf{1.0})$$

Drying shrinkage strain: $\varepsilon_{csd} = k_1 k_4 \varepsilon_{csd.b}$

Concrete

- k_1 is obtained from Figure 3.1.7.2 of AS 3600-2018

- k_4 is equal to 0.7 for an arid environment, 0.65 for an interior environment, 0.6 for a <u>temperate</u> inland environment (Brisbane) and 0.5 for a tropical of near-costal environment.

- $\varepsilon_{csd.b}$ is the basic drying shrinkage strain, given by:

$$\varepsilon_{csd.b} = (\mathbf{0.9} - 0.00\mathbf{5}f_c')\varepsilon^*_{csd.b}$$

$\varepsilon^*_{csd.b}$ is the final drying basic shrinkage strain depends on the quality of the local aggregates and shall be taken as 800×10^{-6} for Brisbane.

Concrete

From AS3600-2018

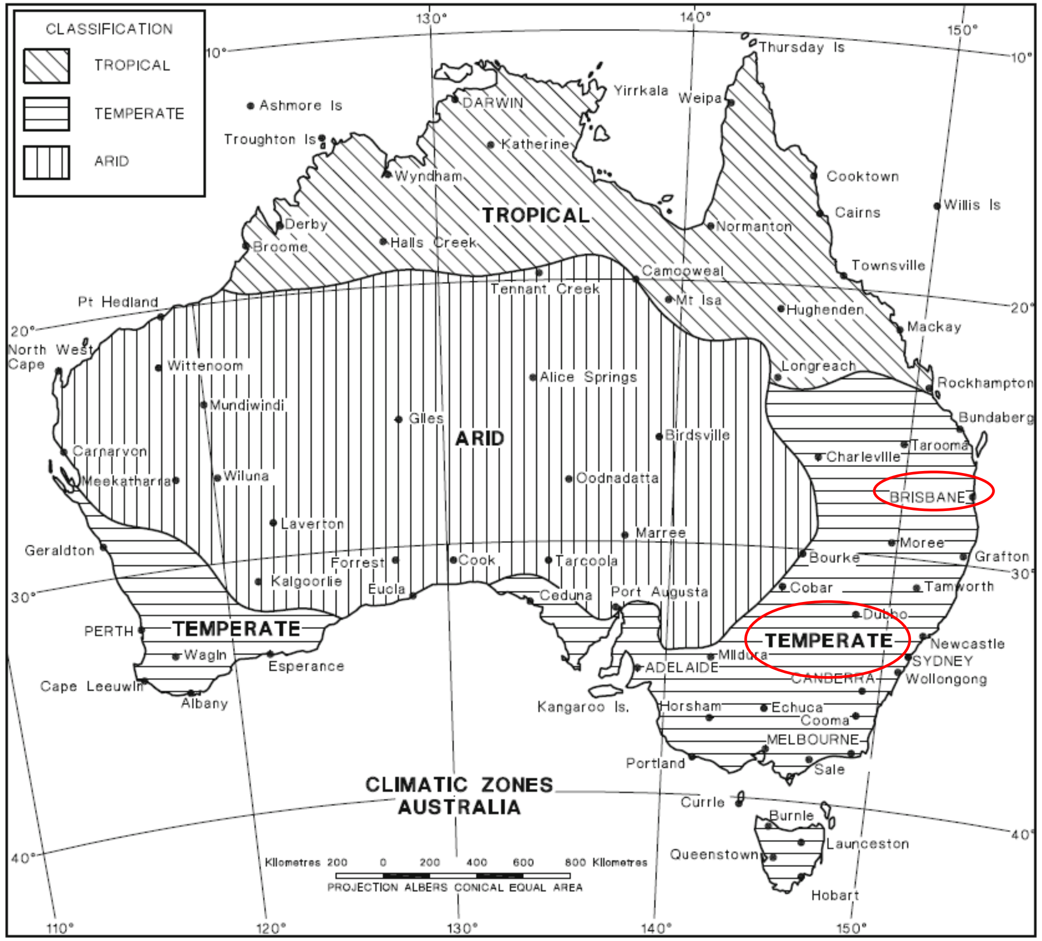
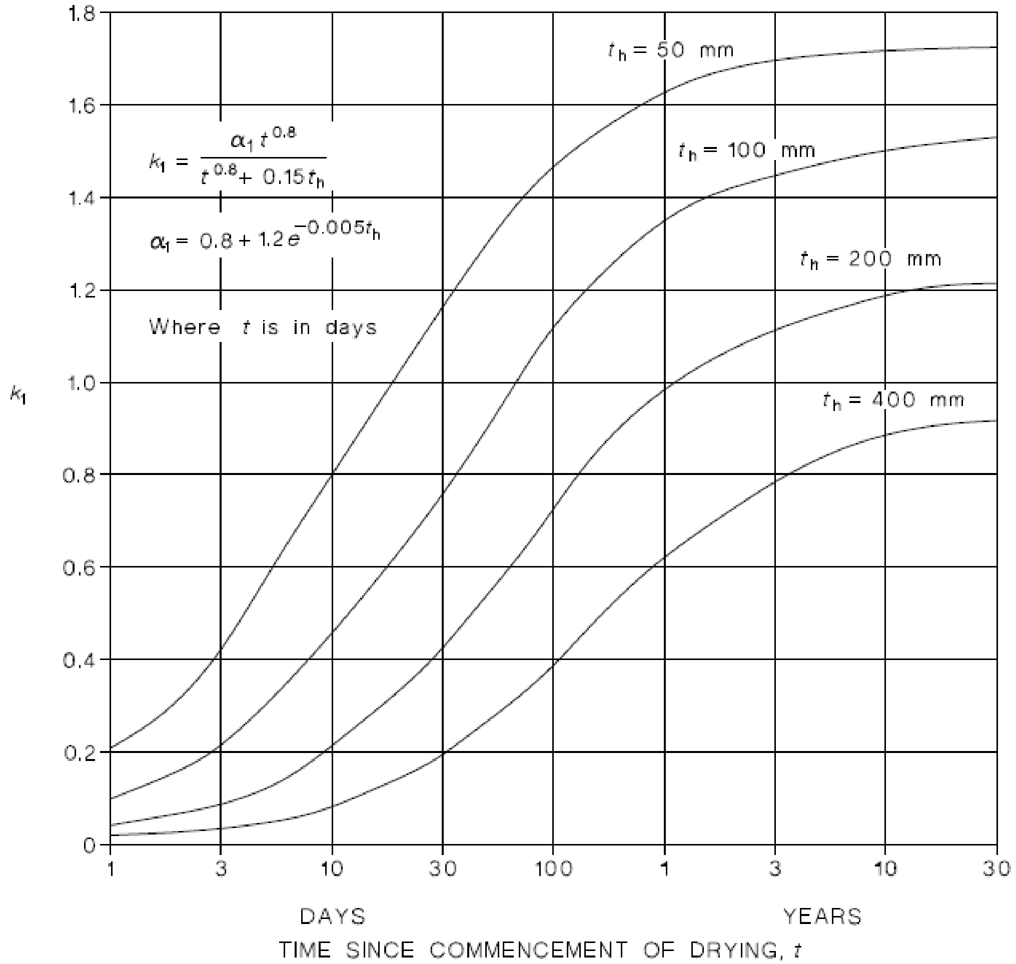


FIGURE 4.3 CLIMATIC ZONES REFERRED TO IN TABLE 4.3

Concrete

- Value k_1



(Note: t_h is a hypothetical thickness given by $t_h = 2A_g/u_e$, where A_g is the gross cross-section area and u_e is the exposed perimeter of a member cross section plus half the perimeter of any closed voids contained therein.)

Concrete

- Final design shrinkage strain after 30 years ($\varepsilon^*_{csd.b}$ = 1000×10^{-6})

TABLE 3.1.7.2

TYPICAL FINAL DESIGN SHRINKAGE STRAINS AFTER 30 YEARS

f'_c (MPa)	Final design shrinkage strain ε^*_{cs} $(\times 10^{-6})$															
	Arid environment				Interior environment				Temperate inland environment				Tropical, near-coastal and coastal environment			
	t_h (mm)				t_h (mm)				t_h (mm)				t_h (mm)			
	50	100	200	400	50	100	200	400	50	100	200	400	50	100	200	400
25	990	870	710	550	920	810	660	510	850	750	610	470	720	630	510	400
32	950	840	680	530	880	780	640	500	820	720	590	460	690	610	500	390
40	890	790	650	510	830	740	610	480	780	690	570	450	660	590	490	390
50	830	740	610	490	770	690	580	460	720	650	540	440	620	550	470	380
65	730	650	560	460	680	620	530	440	640	580	500	410	560	510	440	370
80	630	570	500	420	590	540	480	410	560	520	450	390	500	460	410	360
100	490	460	420	380	480	450	410	370	460	430	400	360	420	400	370	340

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Creep:

- When concrete is subjected to sustained loading, strain increases with time, i.e. concrete exhibits creep.

- The creep strain at any time (*t*) caused by a constant sustained stress (σ_o) shall be calculated from:

$$\varepsilon_{cc} = \frac{\varphi_{cc}\sigma_o}{E_c}$$

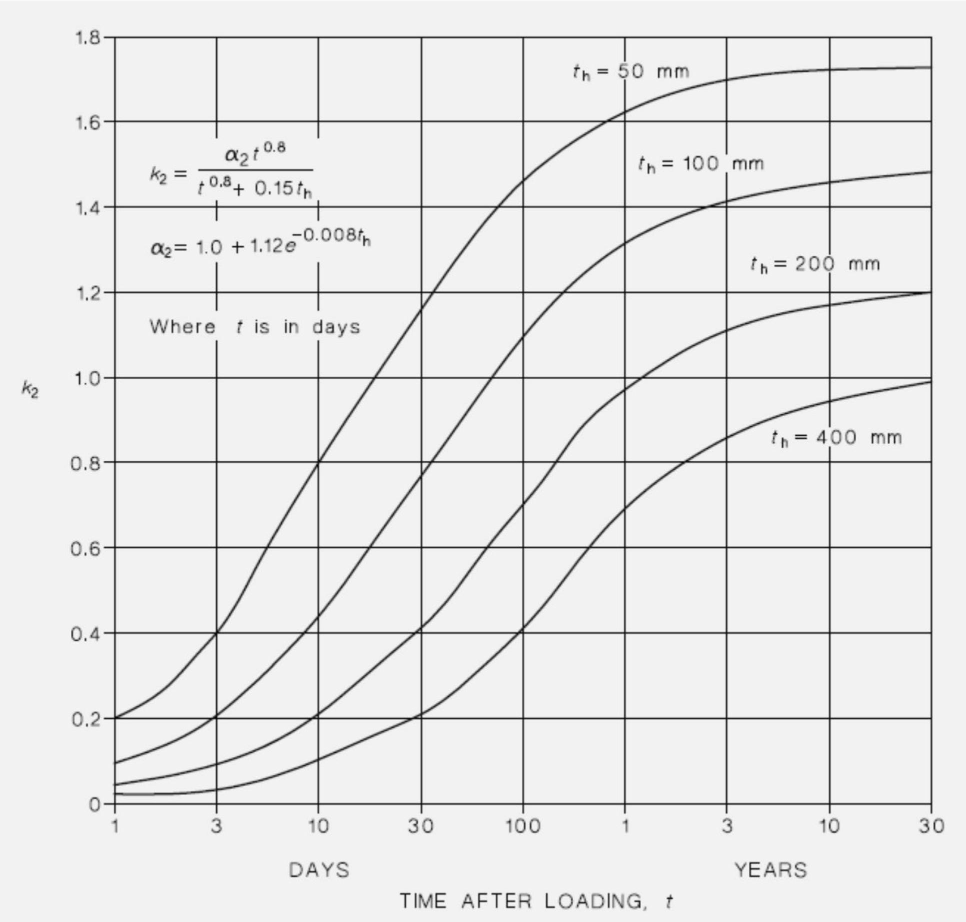
- E_c is the mean modulus of elasticity of concrete at 28 days and φ_{cc} is the design creep coefficient at time *t* determined in accordance with the design creep coefficient.

Concrete

- The design creep coefficient (φ_{cc}) for concrete at any time t, shall be determined from basic creep coefficient ($\varphi_{cc.b}$):

$$\varphi_{cc} = k_2 k_3 k_4 k_5 \boldsymbol{k_6} \varphi_{cc.b}$$

- k_2 is from Fig 3.1.8.3 of AS 3600-2018



Concrete

- k_3 depends on the age of concrete (the loading day)

$$k_3 = \frac{2.7}{1 + \log(\tau)} \text{ for } \tau \geq 1 \text{ day}$$

- k_4 = 0.6 (Brisbane)
- k_5 is a factor accounting for higher strength concrete:

$$k_5 = 1.0 \text{ when } f_c{}' \leq 50 \text{ MPa}$$

$$k_5 = (2.0 - \alpha_3) - 0.02(1.0 - \alpha_3)f_c' \text{ when } f_c' > 50 \text{ MPa}$$

$$\alpha_3 = \frac{0.7}{k_4 \alpha_2}$$

$$k_6 = 1.0 \quad \text{when } \sigma_o \leq 0.45\, f_{cmi}$$

$$k_6 = e^{1.5\left(\frac{\sigma_o}{f_{cmi}} - 0.45\right)} \quad \text{when } \sigma_o > 0.45\, f_{cmi}$$

k_5 decreases as $f_c{}'$ increases

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- $\varphi_{cc.b}$ is the basic creep coefficient which can be obtained from:

TABLE 3.1.8.2

BASIC CREEP COEFFICENT

Characteristic strength (f'_c), MPa	20	25	32	40	50	65	80	100
Basic creep coefficient ($\varphi_{cc.b}$)	5.2	4.2	3.4	2.8	2.4	2.0	1.7	1.5

- The final design creep coefficients (φ^*_{cc}) (after 30 years) for concrete first loaded at 28 days is given in Table 3.1.8.3:

TABLE 3.1.8.3

FINAL CREEP COEFFICIENTS (AFTER 30 YEARS) FOR CONCRETE FIRST LOADED AT 28 DAYS

f'_c (MPa)	Final creep coefficient (φ^*_{cc})											
	Arid environment			Interior environment			Temperate inland environment			Tropical, near-coastal and coastal environment		
	t_h (mm)			t_h (mm)			t_h (mm)			t_h (mm)		
	100	200	400	100	200	400	100	200	400	100	200	400
25	4.82	3.90	3.27	4.48	3.62	3.03	4.13	3.34	2.80	3.44	2.78	2.33
32	3.90	3.15	2.64	3.62	2.93	2.46	3.34	2.70	2.27	2.79	2.25	1.90
40	3.21	2.60	2.18	2.98	2.41	2.02	2.75	2.23	1.87	2.30	1.86	1.56
50	2.75	2.23	1.89	2.56	2.07	1.73	2.36	1.91	1.60	1.97	1.59	1.33
65	2.07	1.75	1.53	1.95	1.66	1.46	1.84	1.59	1.38	1.61	1.38	1.23
80	1.56	1.40	1.29	1.50	1.36	1.25	1.45	1.32	1.22	1.33	1.23	1.14
100	1.15	1.14	1.11	1.15	1.14	1.11	1.15	1.14	1.11	1.15	1.14	1.11

PRESTRESSING TENDONS/WIRES

Prestressing Tendon

- Characteristic minimum breaking strength = f_{pb} (Table 3.3.1 of AS 3600-2018).

- Prestressing steel will not fail when it first reaches the breaking strength. Since steel is ductile, it can still sustain large deformation at the breaking strength.

- Yield strength of tendons = f_{py} shall be taken either as the 0.1% proof stress as specified in AS/NZS 4672.1, or determined by test data. In the absence of test data, $f_{py} \approx 0.8 - 0.83\ f_{pb}$.

- The modulus of elasticity of commonly used tendon (E_p) can be taken as about 200 – 205 GPa (±5 – 10 GPa) depending on the tendon.

- The initial prestressing stress / load is normally about 70% of the breaking strength f_{pb}. This is to ensure elastic behaviour.

Prestressing Tendon

- Tensile strength of commonly used wires, strands and high strength alloy bars:

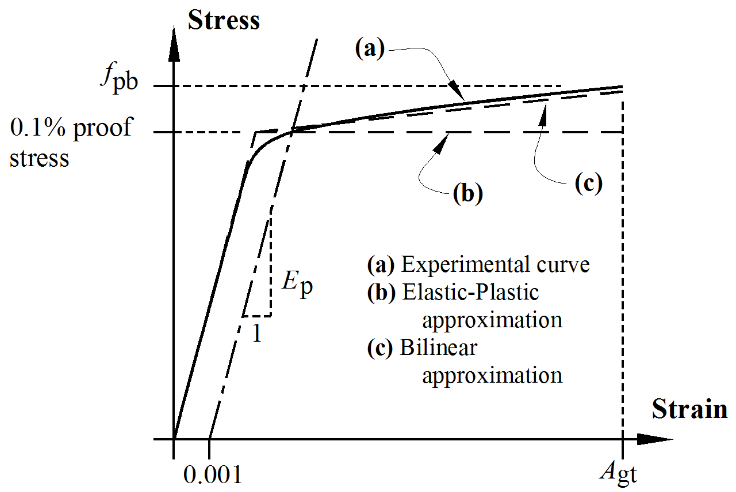


TABLE 3.3.1

TENSILE STRENGTH OF COMMONLY USED WIRE STRAND AND BAR

Material type and Standard	Nominal diameter	Area	Characteristic minimum breaking load	Characteristic minimum breaking strength (f_{pb})
	mm	mm^2	kN	MPa
As-drawn wire, AS/NZS 4672.1	5.0 7.0	19.6 38.5	34.7 64.3	1700 1670
Stress-relieved wire, AS/NZS 4672.1	5.0 7.0	19.9 38.5	33.8 64.3	1700 1670
7 wire ordinary strand, AS/NZS 4672.1	9.5 12.7 15.2 15.2	55.0 98.6 140 143	102 184 250 261	1850 1870 1790 1830
7 wire compacted strand, AS/NZS 4672.1	15.2 18.0	165 223	300 380	1820 1700
Hot-rolled bars, AS/NZS 4672.1 (Super grade only)	26 29 32 36 40 56 75	562 693 840 995 1232 2428 4371	579 714 865 1025 1269 2501 4502	1030 1030 1030 1030 1030 1030 1030

Prestressing Tendon

- The relaxation, which is the reduction of steel stress at sustained load, at any age and stress level, of low-relaxation wire, low relaxation strand and alloy-steel bars should be determined as:

$$R = k_7 k_8 k_9 R_b$$

$$k_7 = \log[5.4(j)^{1/6}]$$

- *j* is the time after prestressing in days
- k_8 is a coefficient depending on the stress in the tendon
- k_9 is a function dependent on the average annual temperature (T) in °C, taken as $T/20 \geq 1.0$
- R_b is a reference value at 20°C at 1000 hours.

Prestressing Tendon

- Coefficient k_8

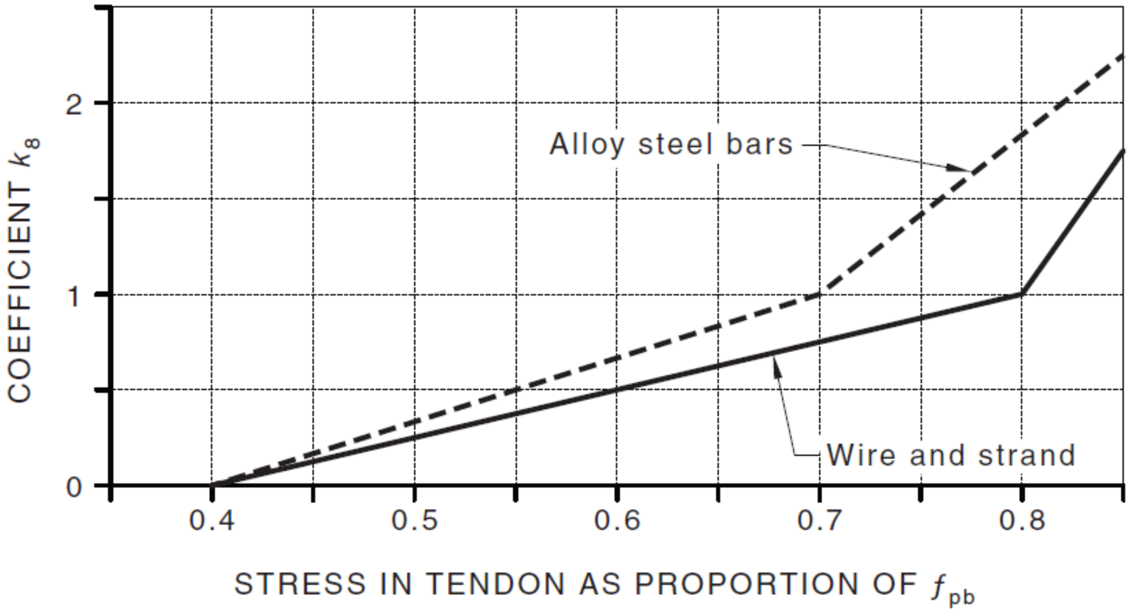
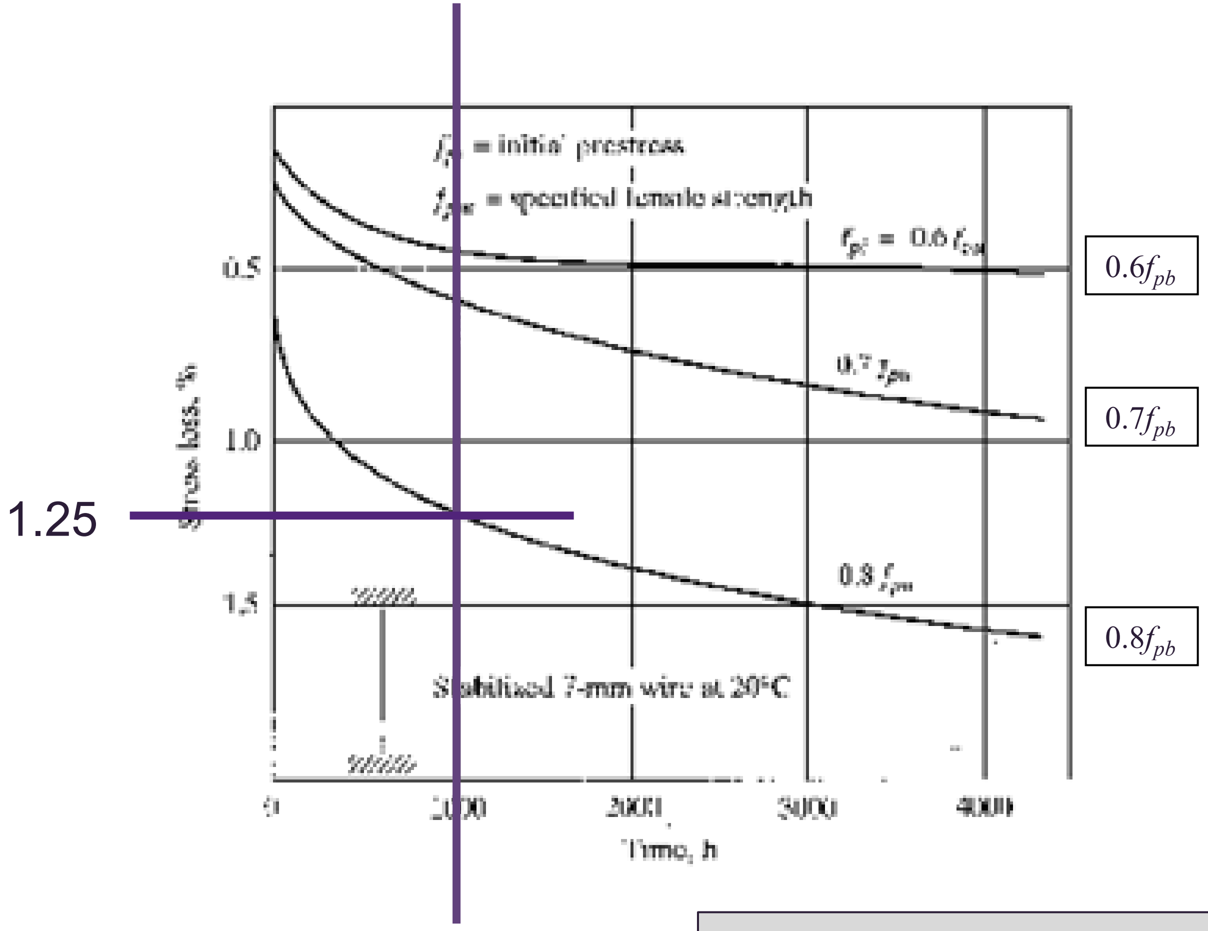


FIGURE 3.3.4.3 COEFFICIENT k_8

Prestressing Tendon

- Some information about R_b



Low relaxation wires, $R_b \approx 2\% f_{pb}$

Prestressing Tendon

<u>Terminology</u>

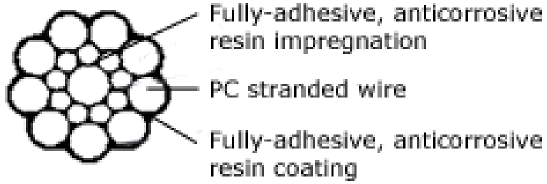
- To describe prestressing steel, we normally refer to wires, strands, cables and/or tendons. What are they?

- Prestressing wire – for post-tensioned construction and occasionally for pre-tensioned work. The individual wires are manufactured by hot-rolling steel billets into round rods. After cooling, the rods are passed through dies to reduce their diameter to the required size.

- The wires are stress-relieved after cold drawing by a continuous heat treatment to produce the prescribed mechanical properties.

Prestressing Tendon

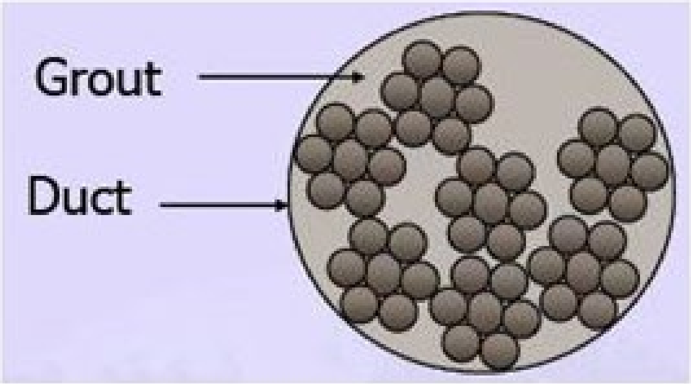
- Strands – almost always used for pre-tensioned members, and is often used for post-tensioned construction.
- Strand is usually fabricated with six wires wound tightly around a seventh of slightly larger diameter. The pitch of the spiral winding is between 12 to 16 times the nominal diameter of the strand.
- The strands are then subjected to a stress-relieving continuous heat treatment to produce the prescribed mechanical properties.



Prestressing Tendon

- Cables – Group of strands normally grouted in a polyethylene sheathing / duct.

- Tendons - A general term describing the prestressing steel, which can be wire or strand / cable.



UNTENSIONED (NON-PRESTRESSING) REINFORCEMENT

Untensioned (Non-prestressing) Reinforcement

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- The characteristic yield strength (f_{sy}) of untensioned (or non-prestressing) reinforcement shall be taken as not greater than the value specified in Table 3.2.1 of AS 3600-2018.

YIELD STRENGTH AND DUCTILITY CLASS OF REINFORCEMENT

Reinforcement		Characteristic yield strength (f_{sy}) MPa	Uniform strain (ε_{su})	Ductility Class
Type	Designation grade			
Bar plain to AS/NZS 4671	R250N	250	0.05	N
Bar deformed to AS/NZS 4671	D500L (fitments only) D500N	500 500	0.015 0.05	L N
Welded wire mesh, plain, deformed or indented to AS/NZS 4671	D500L	500	0.015	L
	D500N	500	0.05	N

NOTE: Reference should be made to AS/NZS 4671 for explanation to designations applying to 500 MPa steels.

- Generally, two types of reinforcement:
  - Plain bar manufactured to AS/NZS 4671 ($f_{sy} \geq 250$ MPa)
  - Deformed bar manufactured to AS/NZS 4671 ($f_{sy} \geq 500$ MPa)

Untensioned (Non-prestressing) Reinforcement

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- The uniform strain (ε_{su}), which is the maximum strain where the uniform yield stress can be up to, is:
  - 0.05 for plain bars
  - 0.015 for low ductility class (L) deformed bars
  - 0.05 for normal ductility class (N) deformed bars

- The modulus of elasticity (E_s) of untensioned reinforcement for all stress values not greater than f_{sy} can be taken as 200,000 MPa.

Summary

- Concrete short-term: Strength and Elastic Modulus
- Stress limits of concrete for design: Transfer and service
- Concrete long-term: Shrinkage and Creep
- Prestressing steel short-term: Strength and Elastic Modulus
- Prestressing steel long-term: Relaxation
- Wires / Strands / Cables / Tendons
- Untensioned (Non-prestressing reinforcement) short-term only: Strength and Elastic Modulus

Flexural Behaviour of uncracked members

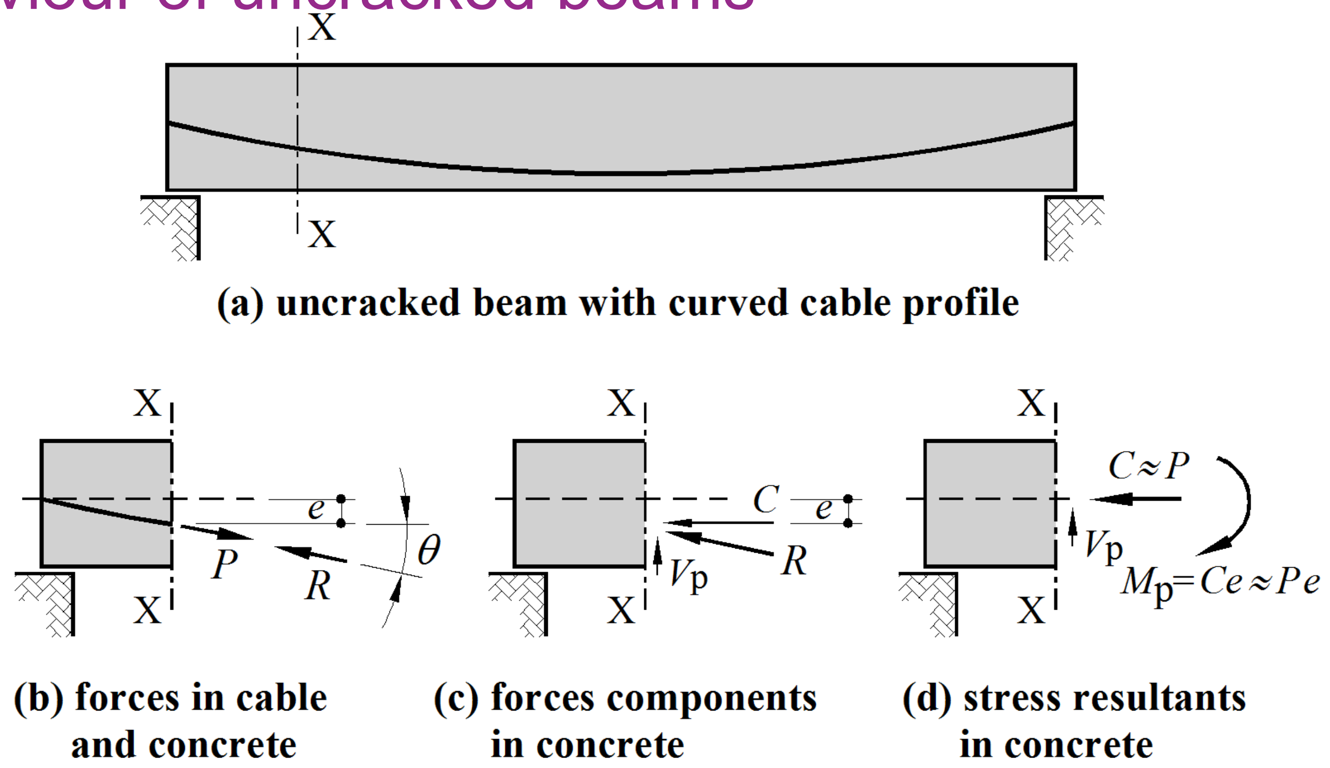
Flexural Behaviour of uncracked members

The short-term of uncracked prestressed concrete beams under working load conditions.

Equivalent loads and load balancing are important design concepts.

Methods for calculating stresses due to prestress and applied loads are presented

Flexural Behaviour of uncracked beams

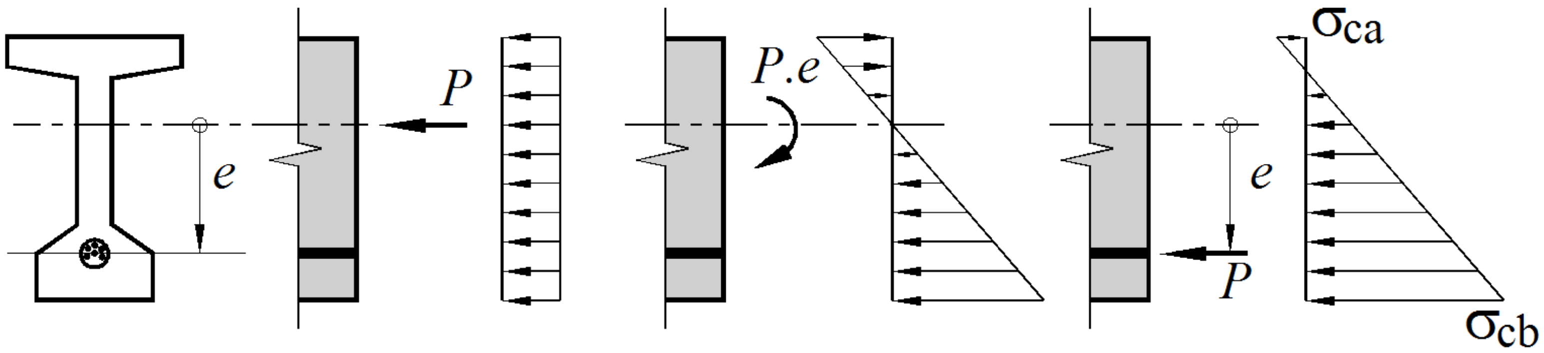


Stress resultants in concrete due to prestress

With *P*=*C*. the concrete compressive stress at any depth *y* below the centroidal axis given by

$$\sigma_{cy} = \frac{P}{A_g} + \frac{Pe}{I_g} y$$

Stresses in a beam section due to prestress – ignoring all external loads and self weight



(a) Stresses due to axial force *P* (b) Stresses due to *Pe* (c) Stresses due to eccentric prestress

Stress at the bottom fibre of the beam after prestress

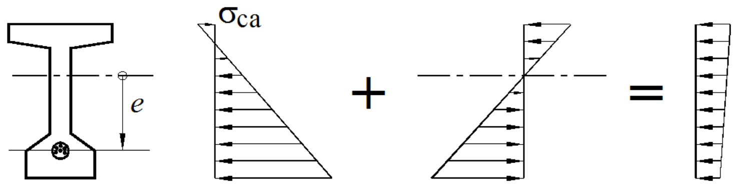
$$\sigma_{cb} = \frac{P}{A_g} + \frac{Pe}{I_g} y_b$$

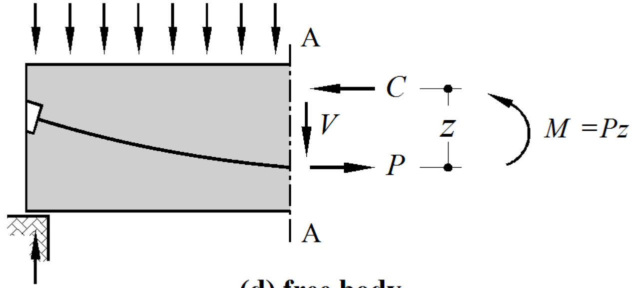
Stress at the top fibre of the beam after prestress

$$\sigma_{ca} = \frac{P}{A_g} - \frac{Pe}{I_g} y_t$$

P= prestress, A = cross sectional area, e= Eccentricity of prestress, ight, I= second moment of area, y_b and y_t distance from NA to the top or bottom edge of the beam

Stresses in a beam section due to prestress plus applied loads (applied moment)





(d) free body

Positive moment M in section due to applied loads produces tensile stress in lower fibres of concrete. Elastic concrete stress at depth y below centroidal axis is given by $\sigma = -My/I_g$ *tension taken as negative*

When M is applied, resultant compressive stress due to prestress is reduced

Resultant compressive C force in concrete acts at distance z above "line of action" of the tensile force P

Horizontal forces in section, P and C are equal and comprise a couple which is equal to applied M:

$$M = Cz = Pz$$

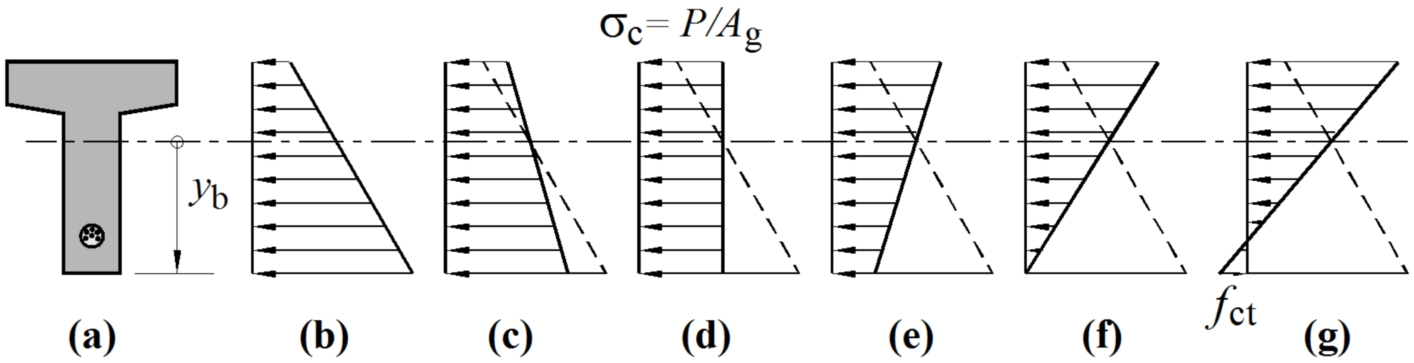
As M increases, P an C remain constant, "line of action" of C moves up in section, and z increases as M increases

Stresses in a beam section due to prestress plus applied loads (applied moment)

C and *P* remain constant, increase in applied *M* accommodate by increase in *z*

As applied moment *M* increases, the concrete compressive stress block changes shape

Concrete stress at level of the centroid does not change $\sigma_c = P/A_g$ (Stress line rotates about centroid)



b) $M = 0$

d) Uniform compressive stress $M = M_p = Pe$ (balanced)

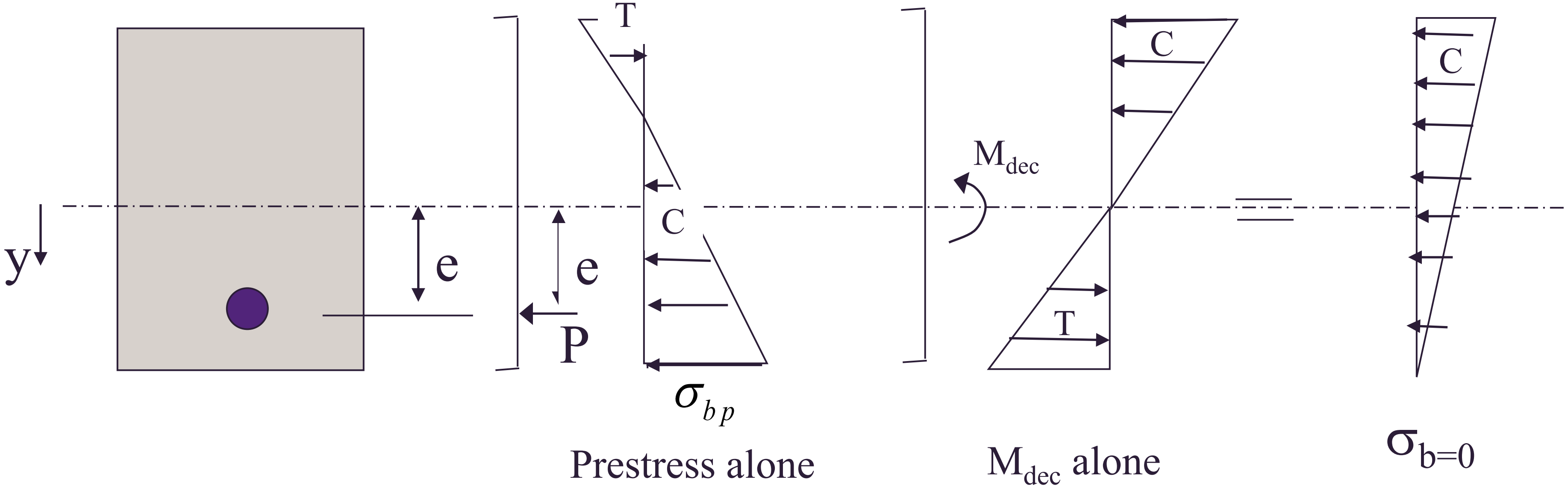
f) Stress is bottom fibre equal to zero, called decompression

g) Final stage reaches concrete tensile strength f_{ct} (cracked section analysis week 3)

Decompression moment

Applied moment at which the bottom fibre stress of the beam becomes zero

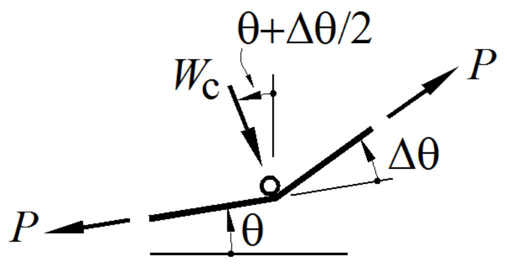
$$M_{dec} = \frac{\sigma_{cbp} I_g}{y_b}$$



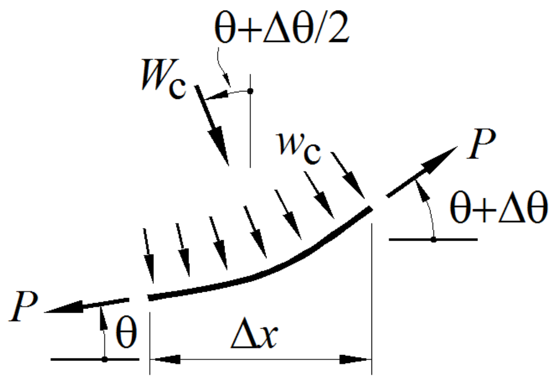
Equivalent Load Concept in Prestressed Concrete

A stressed tendon exerts an equilibrating system of forces on a prestressed beam. The forces act at each end *P*, where the tendon is anchored, and also along the beam wherever the tendon changes direction.

While the tendon exerts a local force on the concrete , the concrete in turn exerts an equal and opposite force on the tendon W_c, which allows the tendon (which has a low flexural stiffness) to change direction. *A force exerted on the beam by the tendon can be regarded as an equivalent load, W_p.*



(a) Kink in tendon



(b) Curve in tendon

The stresses in a beam due to the prestress can be determined by considering the effect of these equivalent loads.

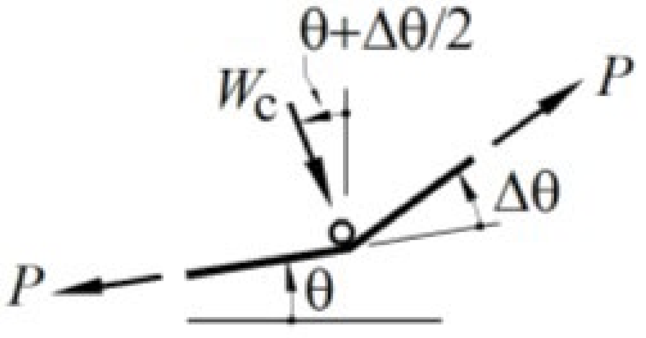
We thus have an alternative method to the one explained above in week 1 lecture for determining the stresses in a beam due to prestress.

The concept of equivalent loads gives us a simple and clear view of the effects of prestress on both statically determinate and indeterminate beams.

It is also the basis for a design concept called load balancing

Equivalent load at a kink in the tendon

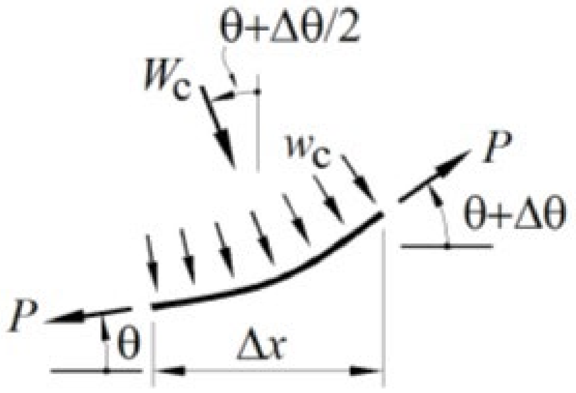
- The slope of the tendon is θ just before the kink and $\theta + \Delta\theta$ just after the kink.
- The tensile force in the tendon on either side of the kink is ***P*** and these are equilibrated by a transverse force W_c, which the concrete exerts though the pin on the tendon.
- The tendon, in turn, exerts a force of W_p through the pin on to the concrete which is, of course, equal and opposite to W_c.
- For equilibrium in the transverse direction we have a force of $W_c = 2P\sin(\Delta\theta/2)$.
- The angle θ and the kink angle $\Delta\theta$ are small, so that we can set $\sin\theta = \tan\theta = \theta$ and hence write:
- $W_p = W_c = P\,\Delta\theta$
- Can assume that the line of action of the equivalent load ***Wp*** is vertical



(a) Kink in tendon

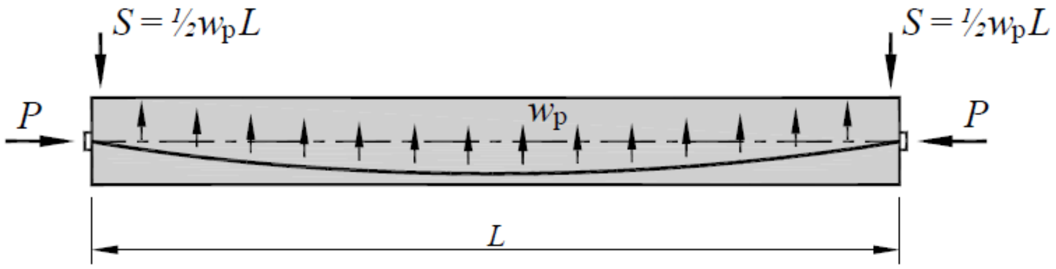
Equivalent load in a curved region of tendon

- a short length of tendon is curved, and a change in slope $\Delta\theta$ occurs over the length $\boldsymbol{\Delta x}$.
- The transverse force acting on the concrete is distributed and has the value $\boldsymbol{w_p}$ kN/m, which is, equal and opposite to $\boldsymbol{w_c}$.
- The local stresses are much smaller and there is no danger of local crushing of the concrete.
- The resultant force $\boldsymbol{W_p}$ acting on the concrete over the length Δx is $\boldsymbol{W_p} = \boldsymbol{W_c} = \boldsymbol{w_c}\,\boldsymbol{\Delta x}$.
- The magnitude of $\boldsymbol{w_p}$ depends on the force $\boldsymbol{P}$ and the curvature κ in the cable, $\boldsymbol{w_p} = \boldsymbol{P} \times \kappa$. If the tendon has a parabolic shape then $\boldsymbol{w_p}$ is uniformly distributed. .

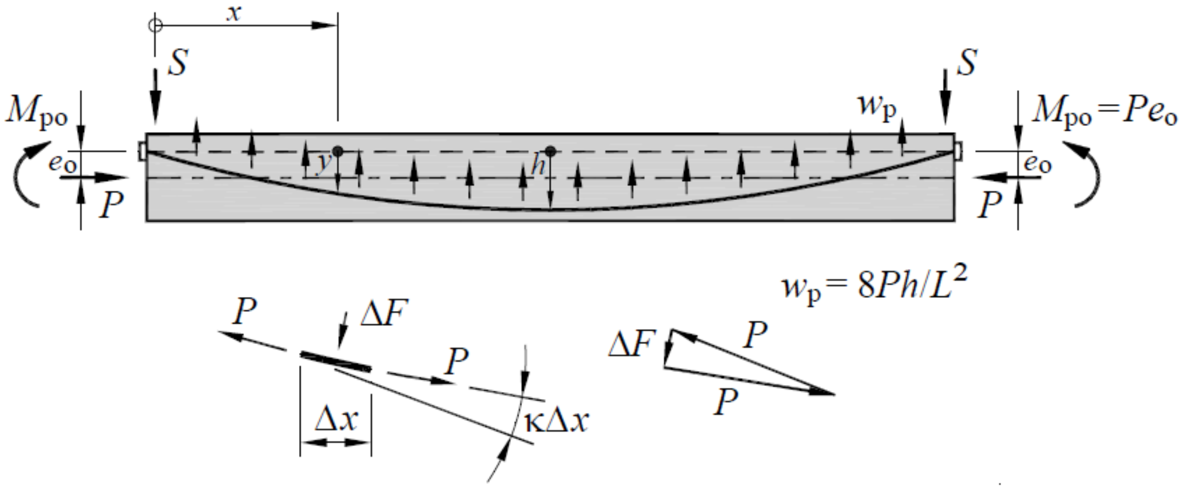


(b) Curve in tendon

Equivalent distributed load for a tendon of parabolic shape



(a) Parabolic cable with zero end eccentricity



(b) Parabolic cable with non-zero end eccentricity

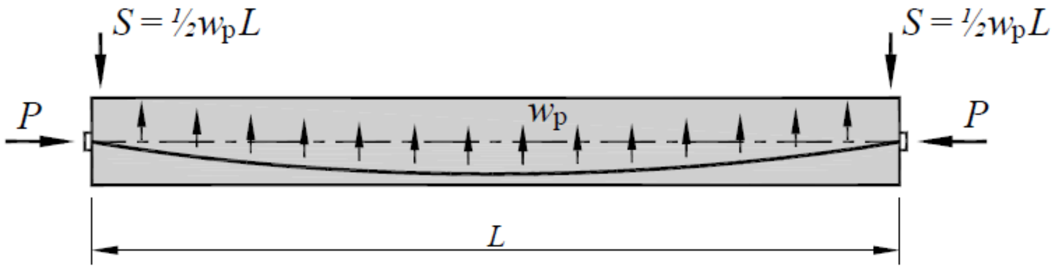
The difference between mid-span eccentric and the mean of the end eccentricities of a parabolic cable is called the sag, and is denoted as ***h***.

The eccentricity ***e*** at any cross-section is a function of the distance ***x*** along the beam . The slope of the tendon is the first derivation of ***e(x)*** and thus:

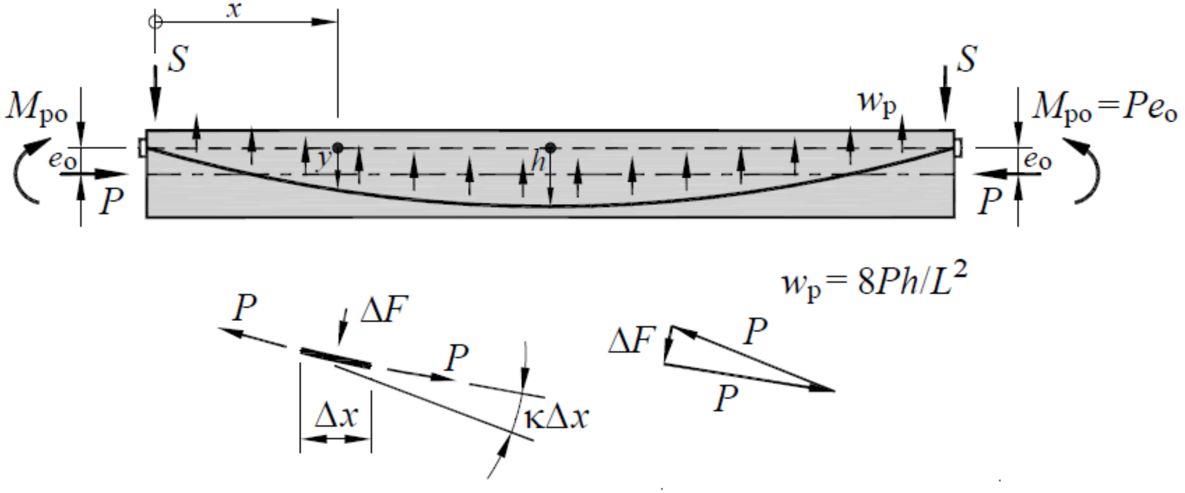
The equation for the parabola can be established based on a generic expression: $\boldsymbol{e = ax^2 + bx + c}$
Boundary conditions: x=0, e=0, e=h and x=L/2 hence

$$e(x) = 4h\left[\frac{x}{L} - \left(\frac{x}{L}\right)^2\right]$$

Equivalent distributed load for a tendon of parabolic shape



(a) Parabolic cable with zero end eccentricity



(b) Parabolic cable with non-zero end eccentricity

$$e(x) = 4h\left[\frac{x}{L} - \left(\frac{x}{L}\right)^2\right]$$

h = difference between mid-span eccentric and the mean of the end eccentricities sag

The slope of the tendon is the first derivation of ***e(x)*** and thus:

$$Slope\theta(x) = \frac{de}{dx} = \frac{4h}{L}\left(1 - \frac{2x}{L}\right)$$

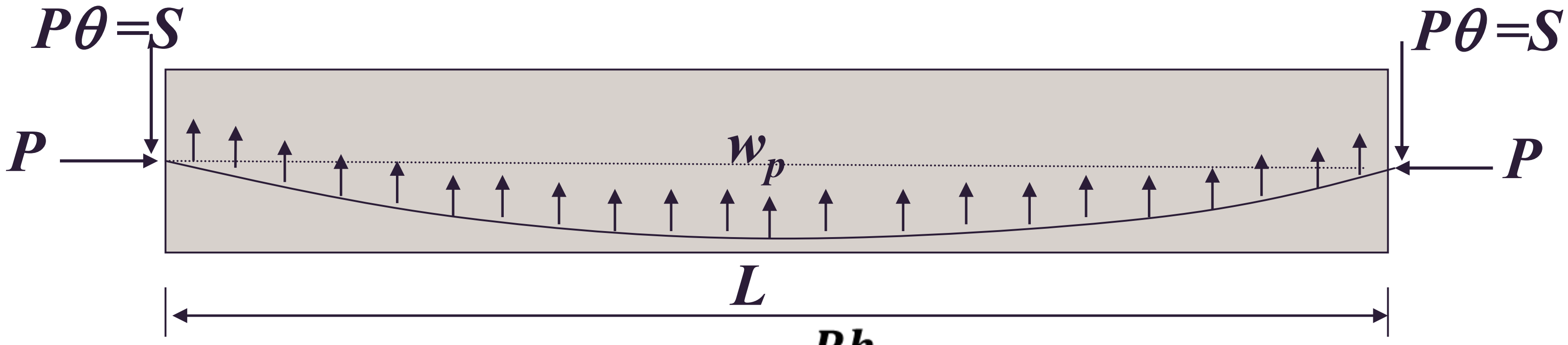
The curvature in the tendon, k, is the rate of change of slope ie. The second derivative of ***e(x)***, which is ***$-8h/L^2$*** and is constant along the span.

A parabolic tendon thus induces a uniformly distributed load of $\boldsymbol{w_p = -8\frac{Ph}{L^2}}$

Equivalent load –Parabolic cable

The slope θ of the tendon at the end of the beam (at the anchorage) is 4h/L,

Transverse end force exerted through the anchorage is ***S = 4Ph/L***, while the horizontal force is taken to be approximately ***P***.



$$w_p = -8\frac{Ph}{L^2}$$

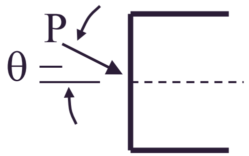
$$Slope\,\theta(x) = \frac{de}{dx} = \frac{4h}{L}\left(1 - \frac{2x}{L}\right)$$

$$Slope\,\theta(x = 0\ and\ x = L) = \frac{4h}{L}$$

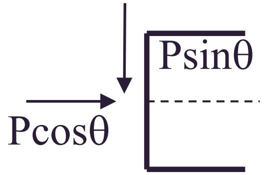
Forces at Anchorages

At its anchorage a prestressing cable exerts on the concrete an equivalent load whose magnitude is that of the prestressing force and whose direction is that of the cable at the anchorage

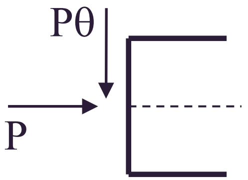
Can use small angle approximations sin $\theta = \theta$, cos $\theta = 1$



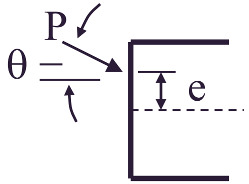
Equivalent load



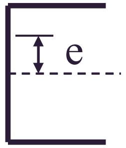
components



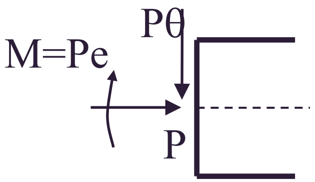
For small angles



equivalent load – eccentric anchorage



components



Equivalent system at centroidal axis

Worked example- Equivalent load Method

The equivalent load method is used in this example to calculate the stresses due to prestress and full load for the beam in lecture example 1 (ex 4.1 from text)

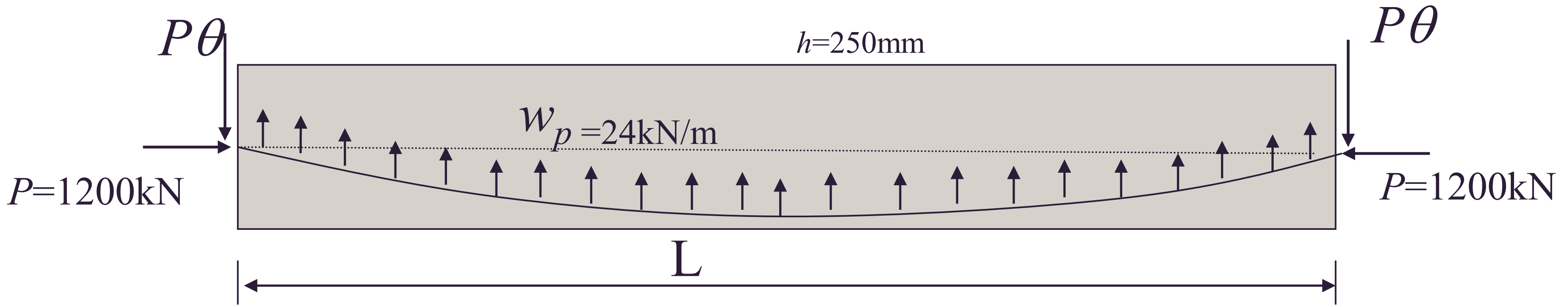
A prestressed concrete beam spans 10 m and has a rectangular cross section 800 mm deep and 400 mm wide. It is post-tensioned by a cable having a cross-sectional area of 1000 mm^2. The cable eccentricity varies parabolically from zero at each end to a maximum of 250 mm at midspan, where the prestressing force is 1200 kN. f'_c = 32 MPa

Determine

The stresses at mid-span due to prestress acting alone

The stress at midspan due to prestress, self weight and a 30 kN/m load

Calculation of stresses using the equivalent load method



$\boldsymbol{P}$ = 1200kN, $\boldsymbol{\theta}$ (x=0 and x= L) = $\boldsymbol{4h/L}$ = (4 x 0.25)/10 = 0.1 rad

↑ Udl due to prestress:

$\boldsymbol{w_p = -\,8Ph/L^2}$ = -(8 x 1200 x 0.25)/ 10^2 = 24kN/m

Downward force at each anchorage (end of tendon) : $\boldsymbol{S = P\theta}$ = 0.1 x 1200 = 120kN

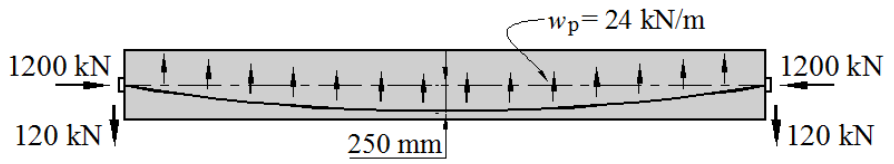
(statistical check $\boldsymbol{2S = w_pL}$ = 240kN

The stress resultants at mid-span due to equivalent loads are:

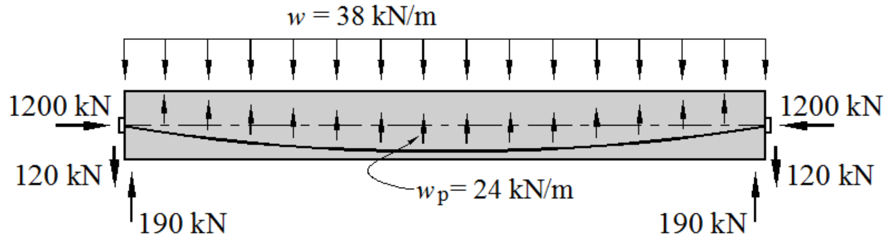
At mid span, the moment due to equivalent loads is : M = -(120 x 5) +(24 x 5 x 2.5) = -300kNm

Axial Force N = 1200kN (Compression)

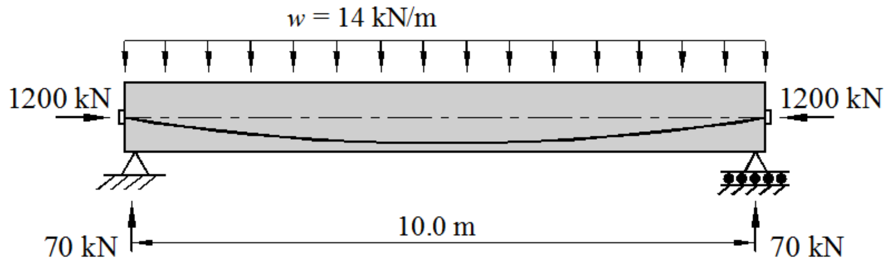
Equivalent loads



(a) Prestress only



(b) Prestress plus applied loads



(c) Resultant loads

The extreme fibre stresses due to this moment are :

$$-\sigma_{tp} = \sigma_{bp} = \frac{My}{I_g} = \frac{300x10^6x400}{17.07x10^9} = 7.03MPa$$

The equivalent axial force produce a uniform compressive stress of:

$$\sigma_c = \frac{1200x10^3}{800x\,400} = 3.75MPa$$

The total extreme fibre stresses due to prestress only *(Fig a above)* are thus:

σ_{tp} =+3.75 -7.03= -3.28 MPa (tension)

σ_{tp} =+3.75 +7.03 = +10.28MPa (compression)

The extreme fibre stresses due to prestress, self weight and live load:

To determine conditions under the full load, we add the applied loads, 30 +8 = 38kN/m to the equivalent upward load of 24kN/m exerted by the prestressing cable.

That is: Net load can be calculated as:

Self weight = 8kN/m$\downarrow$

Live load = 30kN/m $\downarrow$

Prestress equivalent load =w_p =24kN/m $\uparrow$

Net load = 8+30-24=14kN/m $\downarrow$

At mid–span the bending moment is M = $wl^2/8$ = $(14 \text{ x } 10^2)/8$=175kNm

This moment induces extreme fibre stresses of

$$\sigma_{tp} = -\sigma_{bp} = \frac{My}{I_g} = \frac{175\text{x}10^6\text{x}400}{17.07\text{x}10^9} = 4.1\text{MPa}$$

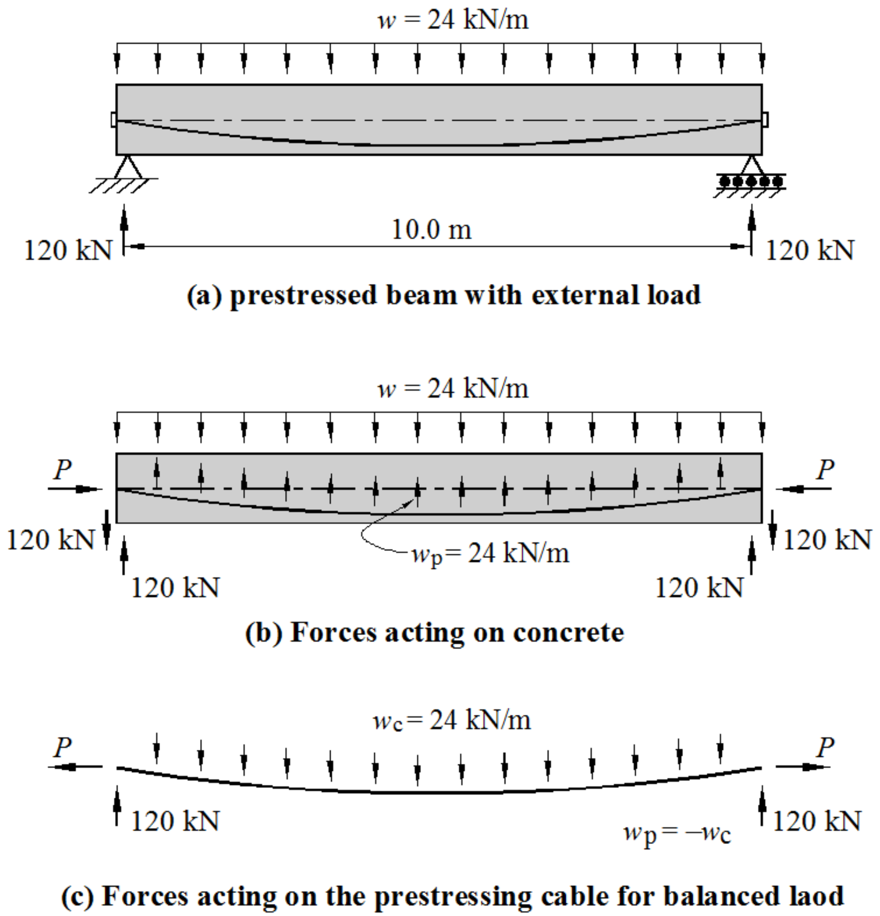
As before, the uniform compressive force due to axial load is 3.75MPa and the final stresses are:

σ_{tp} =+3.75 +4.10= +7.85 MPa (compression)

σ_{tp} =+3.75 -4.10 = -0.35MPa (tension)

Load Balancing

Determining the equivalent upward load due to prestress can be useful to determine the required prestress force to balance a given set of applied loads.



Load balancing; internal equivalent loads are equal and opposite to a system of externally applied design loads

Prestressing Force

Up to this point we have simply used prestressing force, P as constant throughout length of beam.

In reality, loss of prestress force in tendon occurs immediately after tendons are stressed and continue to occur throughout the life of prestressed member

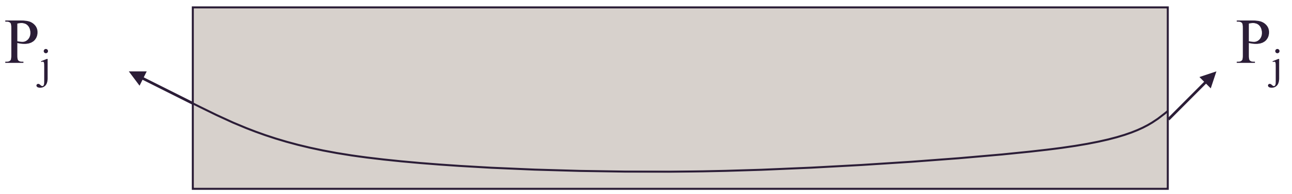
In week 4 you will be shown how to calculate loss of prestress at different stages, namely

Immediate loss – occurs immediately after transfer (includes friction loss)

Deferred loss - Time-dependent takes place gradually over time

Load balancing calculations are based on **effective prestress, P_e** after all losses have occurred.

Calculations of conditions immediately after transfer are based on **initial prestress force, P_i**



Jacking — Transfer — End of design life

$$P_j \xrightarrow[\text{Immediate losses}]{} P_i \xrightarrow[\text{Deferred losses}]{} P_e$$

Worked example- Use of Load balancing to determine prestressing details

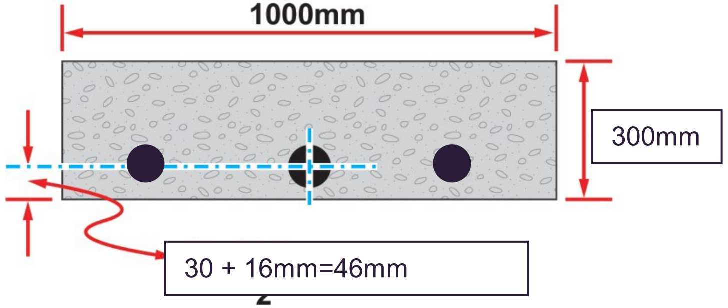
A post-tensioned one-way slab is 300 mm thick and is required to support a superimposed load of 12kPa over a simply supported span of 10 m. Determine the prestress details to keep the slab level under a superimposed live load of 4 kPa.

Assume that the deferred losses of prestress is 18% and the friction loss at mid-span is 5%.

Prestress force required to balance the self-weight.

Load to be balanced = self-weight + 4kN/m

Consider 1 *m* width of the slab and treat it as a beam.



With properties

Area = $\boldsymbol{A_g}$ = 300,000mm^2/m

Second moment of area = $\boldsymbol{I_g}$ = 1000x300^3/12

Self-weight = $\boldsymbol{w_g}$ = 0.3x1x25=7.5kPa (kN/m^2)

Load to be balanced (ie Upward UDL required from prestress)

$\boldsymbol{w_p}$ = 7.5+4.0 = 11.5kPa (kN/m)

The cable eccentricity, ***e*** = (300/2)- 46=104mm, at mid=span sag ***h = e = 0.104m***

$$w_p = \frac{8Ph}{L^2}$$

$$P_e = \frac{w_p L^2}{8h} = \frac{11.5 \times 10^2}{8 \times 0.104} = 1382 kN/m$$

This is the cable force per metre width of slab at mid span after all prestress losses have occured.

Allowing for 18% deferred losses we obtain the initial tendon jacking force at an end as:

$$P_i = \frac{1382}{0.82} = 1685 kN / m$$

- The initial prestressing force at the anchorage, allowing for 5 % losses due to friction is:

$$P_j = \frac{1685}{0.95} = 1774 kN / m$$

- A tendon consisting of four 12.7mm diameter super grade strands *(minimum breaking load per 12.7mm strand is 184kN from table 3.3.1 of AS3600-2009)* stressed to 80 per cent of the minimum breaking stress gives a force of 589kN (=0.80*184*4). The required transverse spacing of these tendons would be

$$s = \frac{1000\text{x}589}{1774} = 337, \text{ie use } 330\text{mm}$$